

Mack Thompson

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EDUCATION

Auburn University

B.S. in Computer Science & Mathematics, Minor in Business Engineering Technology

May 2027

GPA: 3.71/4.00

- Competitive Programming Team, Association for Computing Machinery (ACM), AXA fraternity

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, C#, HTML/CSS

Frameworks & Libraries: React, Flutter, PyTorch, Pandas, Selenium, NumPy, FlaUI

Tools: AWS, Git, Azure, Node, Excel, PowerBI, SQL, Bash, REST, Linux

Awards: ICPC Southeast regional Bronze Medal 2023 & 2023

EXPERIENCE

Software Engineer Intern

Sep 2025 – Present

OCV LLC – Public safety mobile apps

Opelika, AL

- Built analytics dashboards aggregating 250K+ monthly events across 30+ production apps to influence product design.
- Analyzed newly built usage data to reveal low engagement in the wellness apps & led customer discussions, driving the decision to integrate into higher-traffic apps, reducing onboarding effort by 50%.
- Shipped a first-responder wellness app built with SwiftUI through OCV's App Factory on Apple & Google Play stores.

Research Assistant

Feb 2025 – Present

Auburn University – Computer vision & app development

Auburn, AL

- Scraped dermatology dataset of 5k+ labeled images, applying augmentation and ETL preprocessing to stabilize training.
- Uncovered misclassification biases linked to skintone after evaluating EfficientNet-B0 with grouped confusion matrices.
- Led development of a CBT-based mobile therapy app end-to-end as one of two engineers, integrating GPT-powered mental assessments & analysis.

Full-Stack Web Developer

Jul 2025 – Sep 2025

Jup Jewels – e-commerce

Opelika, AL

- Collaborated with the founder to redesign the storefront based on customer behavior analytics; increased traffic by 25%.
- Created a custom jewelry-builder that became the best selling product & generated \$4K revenue in 30 days.

Software Engineer Intern

May 2025 – Aug 2025

Kahua Inc. – PAAS for construction projects

Alpharetta, GA

- Led a team to enable code modification without resetting the test driver, decreasing test-development cycles by 30%.
- Expanded a Azure DevOps testing pipeline by automating 20+ key client workflows, improving deployment reliability.

Founder

Jun 2021 – Dec 2024

Sports Camp

Atlanta, GA

- Scaled to 100+ campers & 6 staff each summer, generating \$100K+ in profit before transferring ownership.
- Deployed AWS Pub/Sub response agent to monitor FAQ, automating 75% of client inquiries & eliminating wait periods.

PROJECTS

Geopolitical Trend Forecaster

Oct 2025

Python, PyTorch, Pandas, Streamlit

[View Repository](#)

- Built ETL pipeline from 200+ years of economic, demographic, military, & trade datasets from open-source archives.
- Developed quantitative scoring algorithms for country strength & geographical advantage to model competitiveness.
- Trained deep temporal models (TCNs) to project geopolitical shifts & relative country standings 30 years into the future.

Social Event Planning Platform

Aug 2024

React, SQL, WebSockets, AWS

[View Repository](#)

- Created a live-messaging & calendar sharing platform with a WebSocket server & responsive React frontend.
- Implemented a RESTful API library, enabling CRUD operations over a PostgreSQL backend for users & interactions.
- Containerized the server and deployed on AWS, implementing environment-variable secrets & auto-rebuild workflows.